

Time 3 Hours

Note: Attempt any two parts from each question. All questions carry equal marks. All part of a question must be attempt at one place.

Q.1 (a) Is java pure object oriented programming? Justify your answer. Explain any five important features of JAVA. 6

(b) Write a Java program to check whether a year is leap or not. 6

(c) What will be the outcome of following code snippet? Also justify your answer. 6

```
(i) public class Code {  
    public static void main(String[] args) {  
        for(int i = 0; i < 1; i++) {  
            System.out.println(i+' ');  
        }  
    }  
}  
  
(ii) public class Code {  
    public static void main(String[] args) {  
        method(null);  
    }  
    public static void method(Object o) {  
        System.out.println("Object method");  
    }  
    public static void method(String s) {  
        System.out.println("String method");  
    }  
}
```

Q.2 (a) Write a program to display all the numbers between an interval which are divisible by 3 and 5. 6

(b) Write a program to display all possible subsets of a string. 6

(c) What will be the outcome of following code snippet? Also justify your answer. 6

```
(i) class Code {  
    public static void main(String args[]) {  
        System.out.println(value());  
    }  
    int value() {  
        return 1;  
    }  
}  
  
(ii) —class Parent {  
    public void Print() {  
        System.out.println("Parent");  
    }  
    —class Child extends Parent {  
        public void Print() {  
            System.out.println("Child");  
        }  
    }  
    class Main {  
        public static void PrintMain(Parent o) {  
            o.Print();  
        }  
        public static void main(String[] args) {  
            Parent x = new Parent();  
            Parent y = new Child();  
            Child z = new Child();  
        }  
    }  
}
```

```
PrintMain(x);  
PrintMain(y);  
PrintMain(z);} }
```

- Q.3 (a) Write a program to check whether the entered matrix is identity matrix or not. 6
- (b) Differentiate between runtime polymorphism and compile time polymorphism. Write a java program to create an abstract class named shape that contains two integers and an empty method named print Area(). Provide three classes named Rectangle, Triangle and Circle such that each one of the classes extends the class shape. Each one of the class contains only the method print Area() that print the area of the given shape. 6
- (c) (i) Why final and abstract can't be used at a time? 6
(ii) Is it necessary for abstract class to have abstract method?
(iii) Can you import same package/class twice? Will the JVM load the package twice at runtime?
- Q.4 (a) Create a user-defined exception named CheckArgument to check the number of arguments passed through the command line. If the number of argument is less than 5, throw the CheckArgumentException, else print the addition of all the five numbers. 6
- (b) Differentiate between *abstract* class and *interface* with example. 6
- (c) Explain thread life cycle. Write a program for yield() and join() method. 6
- Q.5 (a) Explain hierarchy of I/O streams in java. Write a program to illustrate the use of SequenceInputStream. 6
- (b) Write a GUI program to performs multiplication of two numbers using Event Handling. 6
- (c) Write a Java program to read a text file named as "test.txt". Program should count and print the occurrences of the word "at" in the file. Program should not count the occurrence of "at" inside any other word (i.e. "rat"). 6